

Completeness of the invariant-subspace lattice: Conjecture 6825

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Let X be a real or complex topological vector space and let $A : X \rightarrow X$ be a linear operator. Write $\text{Lat } A$ for the closed linear subspaces M with $A(M) \subseteq M$. The standard order is inclusion. No boundedness or continuity assumption on A is needed for the following conclusion. If the algebraic convention omits topological closedness, the same proof applies with every subset declared closed.

Theorem 1. *The partially ordered set $(\text{Lat } A, \subseteq)$ is a complete lattice. For every family $\mathcal{F} \subseteq \text{Lat } A$,*

$$\bigwedge \mathcal{F} = \bigcap_{M \in \mathcal{F}} M, \quad \bigvee \mathcal{F} = \bigcap \{U \in \text{Lat } A : M \subseteq U \text{ for every } M \in \mathcal{F}\}.$$

The intersection over an empty family is X .

Proof. An arbitrary intersection of closed sets is closed. An intersection of linear subspaces contains zero and is closed under addition and scalar multiplication. If x belongs to every member of an invariant family, then so does Ax . Thus the first displayed intersection is an element of $\text{Lat } A$. It is contained in every member of $\mathcal{F}$, and every common lower bound is contained in the intersection; hence it is the greatest lower bound.

The family of upper bounds is nonempty, since it contains X . Its intersection is again a closed invariant linear subspace. Every member of $\mathcal{F}$ is contained in this intersection, because it is contained in each upper bound. Conversely the intersection is contained in each upper bound. It is therefore the least upper bound. This argument includes the empty family, so arbitrary meets and joins exist. Inclusion is reflexive and transitive, and antisymmetry follows from equality of the underlying subsets. These are the complete-lattice axioms in the standard order. $\square$

Meaning of the source. The source's English and Chinese statements concern completeness in the standard order. A complete lattice need not be totally ordered. The result above proves existence of arbitrary infima and suprema; it makes no claim that arbitrary invariant subspaces are comparable.

Formal verification. The self-contained Lean file represents subsets by predicates $X \rightarrow \text{Prop}$ and specifies the usual closed-set axioms of a topology. An invariant subspace carries proofs of closedness, zero membership, closure under the actual addition and scalar multiplication operations, and invariance under the actual map A . The proof needs no vector-space identities and is therefore proved in the stronger generality of arbitrary operations of the same arities. Every real or complex vector space supplies these operations; no numerical or finite-dimensional replacement is made.

Families are arbitrary predicates on the type of invariant subspaces, including infinite and empty families. The closedness proof explicitly converts such a family into a family of its underlying subsets and proves equality of the two intersection predicates. Thus the closed-set axiom is applied to the actual intersection. The carrier of the join is defined as the intersection of all upper bounds. Extensional equality of subspaces, all order axioms, and both universal properties are proved.

The theorem `conjecture6825` bundles these complete-lattice properties. A discrete-topology instance covers the convention of algebraic subspaces. A concrete instance with integers, their usual operations, and $A(n) = 2n$ additionally witnesses consistency of the more general interface. It is not a restriction of the theorem to integers. The project uses Lean 4.19.0 and `Std` only, with no unproved axioms or external decision oracle; printed dependencies are the standard logical axioms `propext` and `Quot.sound`.